

Red-clump angular sizes with intensity interferometry: what `hbtsim` and the NewEra models add to Kim & Kaiser (2026)

P. Nugent (with Claude)

30 September 2026 — draft v2

Summary

Kim & Kaiser (2026, PASP 138, 044202) propose intensity interferometry (II) as an independent check of the VLTI/PIONIER angular diameters that calibrate the red-clump surface-brightness–colour relation, and so the 1% distance to the LMC. For HD 17652 (β For, $V = 4.46$, $\theta_{\text{LD}} = 1.835$ mas) with two 4 m telescopes they find a fractional scale precision $\sigma_s < 0.007$ in a 2 h H-band exposure at ~ 100 m; for HD 360 ($V = 5.99$, 0.906 mas) $\sigma_s < 0.03$. We reproduced these numbers on `hbtsim`’s photon budget and asked what the package and the angle-resolved NewEra PHOENIX models add. Four results:

1. Their noise formula is `hbtsim`’s matched filter with $p_2 = \frac{1}{2}$ (unpolarized light), to 5×10^{-8} . With the model spectrum anchored to their photometry, their Table 1 (the zero-baseline S/N per band) reproduces to 1–6%, and their H-band result to within its rounding ($\sigma_s = 0.0068$ at 120 m for HD 17652 against “ < 0.007 ”; 0.027 for HD 360 against “ < 0.03 ”).
2. The *radius convention* is the largest systematic. In a spherical model of a giant the Rosseland $\tau = 1$ radius, the visible limb and the outer boundary differ; for the 5000 K, $\log g = 0$ model the outer boundary is 16.5% beyond $\tau = 1$ and the visible limb is 13.5% outside it. The 4800 K, $\log g = 2.5$ giant will sit between that and the dwarf’s 0.14%, plausibly at a few per cent — comparable to the 1% goal.
3. Spectral multiplexing in the optical with *real* detectors is at parity with the single H filter to $\times 4$ better in time (EON-SII’s 1000-channel SPAD: $\sigma_s = 0.0035$ in 2 h at 40 m); their “factor ~ 100 ” holds only for an ideal detector (unit PDE, 16.6 ps jitter over 400–950 nm), and even then it is $\times 45$.
4. Multiplexing measures the limb-darkening chromaticity the authors worry about: $\theta_{\text{UD}}/\theta_{\text{LD}}$ runs from 0.89 at 400 nm to 0.955 at 950 nm, a $\sim 3\sigma$ slope per 2 h on HD 17652.

The red-clump model does not exist yet; every profile-dependent number uses labelled standards that bracket the gravity (Section 3).

1 Set-up

Stars. HD 17652 (G9 IIIb; $T_{\text{eff}} = 4786$ K, $\log g = 2.5$, $\theta_{\text{LD}} = 1.835$ mas from Gallenne et al.; $d = 54.2$ pc) and HD 360 (K1 II; 4764 K, 2.5, 0.906 mas; 111 pc), as `hbtsim.single.SingleStar` objects. The model spectrum is anchored to the V , H and K magnitudes the paper adopts (4.456,

2.256, 2.139 and 5.986, 3.757, 3.653; Vega $\rightarrow$ AB offsets 0.02, 1.39, 1.85), the offset interpolated in $\log \lambda$ between the anchors. Unanchored, the stand-in models are 0.2 mag too bright in V and H .

Instrument. Their fiducial: two 4 m telescopes, 0.3 overall throughput, $\sigma_t = 16.6$ ps per detector (their formula uses σ_t ; the “42.4 ps FWHM” quoted alongside would be 18.0 ps), no dark counts or readout limit (`diameter.KK_TELESCOPE`, `KK_DETECTOR`); single top-hat filters $BVRHK$. For the multiplexed cases the telescopes are the same and the detectors are `hbtsim`’s presets: the SPAD Lambda (320 channels, PDE 22–49%, 120 ps FWHM, 10^8 cps time-tag link or an assumed correlator readout), an $R = 5000$ SPAD Lambda correlator backend (4325 channels), and EON-SII’s QUASAR SPAD (1000 channels over 400–550 nm, 20 ps, 1 GHz link). Pupil averaging over the 4 m apertures is on; the uniform-disk inversion inverts the smeared model.

Observable. As in the paper, the scale s stretches a model profile, $I(\theta; s) = I_0(\theta/s)$, and $\sigma_s^{-2} = \sum_{\text{ch}} (\partial|V|^2/\partial \ln s)^2 / \sigma^2(|V|^2)$ over every channel, with $\sigma(|V|^2)$ from the matched-filter photon budget (`hbtsim.diameter.scale_precision`).

Models. NewEra angle-resolved tables (HSR-RF, $[M/H] = 0$) binned to 0.02 nm over 380–1000 nm and to 0.1 nm over 380–2500 nm. **No model near 4800 K / $\log g = 2.5$ exists in the Hamburg share:** the cool box holds $\log g \geq 4.0$ at these temperatures, and $\log g = 0$ –2 only below 4000 K, plus a single 5000 K / $\log g = 0$ supergiant. Stand-ins: the 4800 K / $\log g = 4.5$ dwarf (right temperature, wrong gravity) and the 5000 K / $\log g = 0$ supergiant (the other side of the gravity bracket).

2 Results

2.1 Comparison with the paper

Their Eq. 7, $\sigma(|V|^2)^{-1} = (d\Gamma/d\nu)(T/\sigma_t)^{1/2}(128\pi)^{-1/4}$, is exactly `snr.g2_snr` with $p_2 = \frac{1}{2}$ and a Gaussian pair kernel of width $\sqrt{2}\sigma_t$; the test suite checks the two agree. Their Table 1 lists this inverse noise at zero baseline per band in 2 h; Table 1 sets it against `hbtsim` with the anchored model (`scripts/redclump_compare.py`).

Table 1: Zero-baseline $1/\sigma(|V|^2)$ in 2 h on their instrument: their Table 1 against `hbtsim` (matched filter, model spectrum anchored to their VHK photometry; ratio in parentheses) and the unanchored dwarf stand-in. R and I differ because they interpolate fluxes between V , H , K and Gaia while `hbtsim` interpolates the anchor offset over the model spectrum.

	V	R	I	H	K
HD 17652, paper	28.9	40.4	69.7	188.5	168.6
hbtsim, anchored	28.6 (0.99)	45.4 (1.13)	66.4 (0.95)	181.8 (0.96)	178.1 (1.06)
hbtsim, unanchored stand-in	34.3	54.8	80.4	224.1	185.6
HD 360, paper	7.07	9.81	17.0	47.3	41.8
hbtsim, anchored	6.99 (0.99)	11.2 (1.14)	16.4 (0.96)	45.6 (0.96)	44.2 (1.06)
hbtsim, unanchored stand-in	8.37	13.4	19.6	54.6	45.2

The photon budgets agree to a few per cent band by band, and the H-band σ_s agree within their rounding. The remaining differences are inputs, not physics: the R/I flux interpolation,

Table 2: Fractional scale precision σ_s in 2 h with the paper’s instrument, one filter, baseline scanned 20–300 m (anchored fluxes, dwarf stand-in profile; the supergiant profile changes these by $< 1\%$). Optimum baselines in parentheses. The paper quotes < 0.007 at ~ 100 m (HD 17652) and < 0.03 (HD 360) in H , “ $\lesssim 0.01$ at ~ 150 m” in H and K , and optical uncertainties “2–10 times larger” than H (here $3.3\text{--}6.4\times$).

star	V	R	I	H	K
HD 17652	0.044 (45 m)	0.032 (50 m)	0.023 (60 m)	0.0068 (120 m)	0.0069 (160 m)
HD 360	0.18 (85 m)	0.13 (100 m)	0.092 (125 m)	0.027 (245 m)	0.028 (300 m)

the K filter’s effective wavelength, and the Vega-to-AB offsets. The optimum sits at $|V|^2 \approx 0.33$ ($x = \pi\theta B/\lambda \approx 2.0$) in every band; for HD 360 that is 245 m in H , not 100 m.

2.2 The radius convention

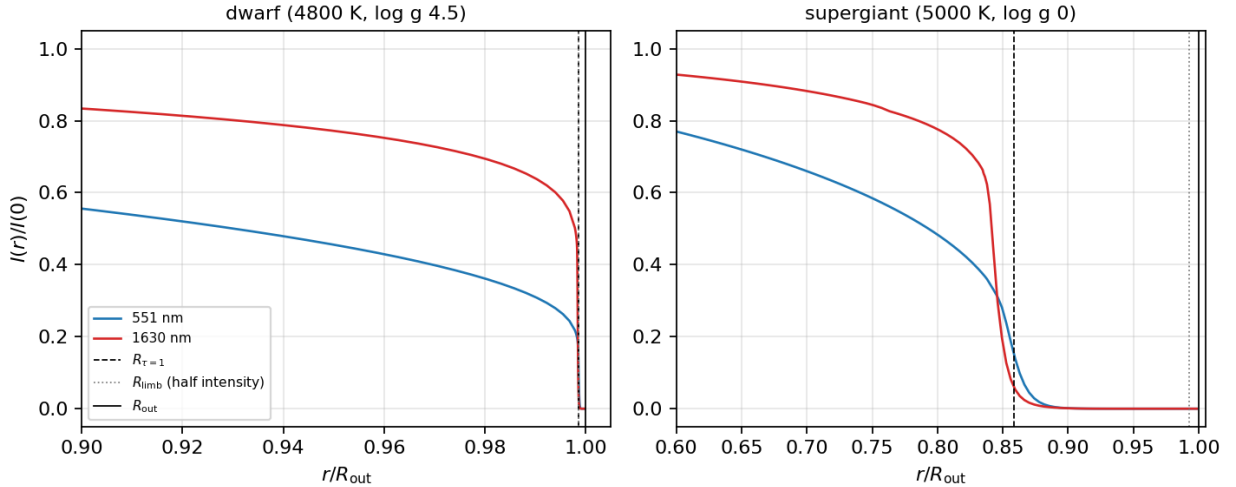


Figure 1: Centre-to-limb intensity of the two stand-in NewEra models at 551 and 1630 nm against radius in units of the model’s outer boundary, with the Rosseland $\tau = 1$ radius (dashed), the apparent limb where the continuum intensity halves (dotted) and the outer boundary (solid). For the dwarf the three coincide to 0.14%; for the supergiant the visible limb lies 13.5% outside $\tau = 1$.

Table 3: Limb geometry and uniform-disk ratios of the stand-in models ($x = 1.5$, 4 m pupils).

model	$R_{\text{out}}/R_{\tau=1}$	$R_{\text{out}}/R_{\text{limb}}$	$\theta_{\text{UD}}/\theta_{\text{LD}}$ (V)	(H)
4800 K, $\log g = 4.5$ (dwarf)	1.0014	1.0014	0.927	0.971
5000 K, $\log g = 0$ (supergiant)	1.165	1.0076	0.924	0.963

The paper’s s multiplies a SATLAS profile normalized to a reference diameter, and its σ_s is compared directly with PIONIER’s limb-darkened diameters. Which radius the reference is matters at the per-cent level for a giant (Fig. 1): the profile shape moves $\theta_{\text{UD}}/\theta_{\text{LD}}$ by only 0.3% (V) to 0.8% (H) between the two extreme gravities, but the definition of “the radius” moves by 16%. An II

validation at 1% must fit the same profile with the same radius definition as the PIONIER analysis, or it compares conventions rather than stars.

2.3 Multiplexing

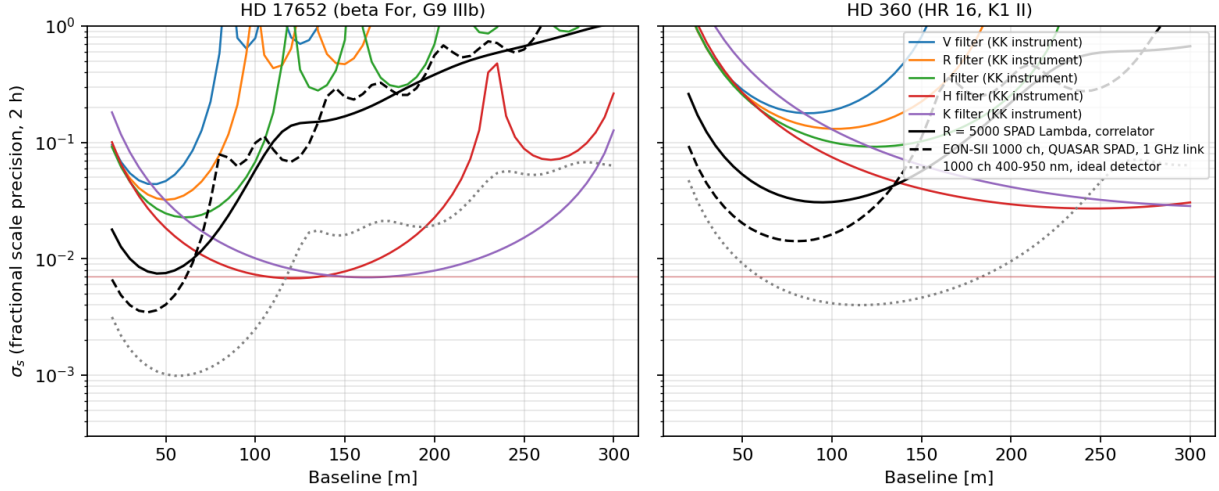


Figure 2: σ_s against baseline in 2 h: the paper’s single filters (colours) and three multiplexed optical backends (black and grey), dwarf stand-in profile. The thin red line is the paper’s $\sigma_s = 0.007$. Beyond ~ 60 m (HD 17652) the optical channels pass through the first null, hence the structure. The grey dotted curve is their ideal detector (unit PDE, 16.6 ps) with 1000 channels: that is the “factor ~ 100 ” in time (here $\times 45$). Real detectors give parity (the $R = 5000$ SPAD Lambda correlator) to $\times 4$ in time (EON-SII).

Table 4: Best σ_s in 2 h and hours to $\sigma_s = 0.007$ (scaling as $1/T$).

backend	HD 17652	HD 360
KK instrument, one H filter	0.0068 (120 m) $\rightarrow$ 1.9 h	0.027 (245 m) $\rightarrow$ 30 h
ideal detector, 1000 ch 400–950 nm	0.0010 (55 m) $\rightarrow$ 0.04 h	0.0040 (115 m) $\rightarrow$ 0.65 h
SPAD Lambda 320 ch, time-tag	0.11 $\rightarrow$ 455 h	0.11 $\rightarrow$ 454 h
SPAD Lambda 320 ch, correlator	0.026 $\rightarrow$ 28 h	0.11 $\rightarrow$ 454 h
$R = 5000$ (4325 ch), correlator	0.0075 (45 m) $\rightarrow$ 2.3 h	0.031 (95 m) $\rightarrow$ 38 h
$R = 5000$, correlator + PBS	0.0053 $\rightarrow$ 1.1 h	0.022 $\rightarrow$ 19 h
EON-SII 1000 ch, QUASAR SPAD, 1 GHz link	0.0035 (40 m) $\rightarrow$ 0.49 h	0.014 (80 m) $\rightarrow$ 8.2 h

Two practical points: the optical channels do their work at 40–120 m, inside the PIONIER-like baselines, whereas HD 360 needs 245 m in H ; and the H-band case itself requires an infrared photon counter with picosecond jitter (InGaAs SPAD, SNSPD) that has not been modelled.

2.4 Chromaticity

The uniform-disk diameter runs 6% across 400–950 nm (0.895, 0.905, 0.919, 0.930, 0.945, 0.955 in the bins 400–450, 450–500, 500–550, 550–650, 650–800, 800–950 nm). The two stand-ins differ by $\leq 1\%$ in the continuum. With $\sigma(\theta_{UD})/\theta = 0.59$ per channel in 2 h on HD 17652, 50 nm bins reach $\sim 3\%$ and the continuum slope is a $\sim 2\sigma$ measurement per 2 h ($\sim 5\sigma$ per night): the limb darkening

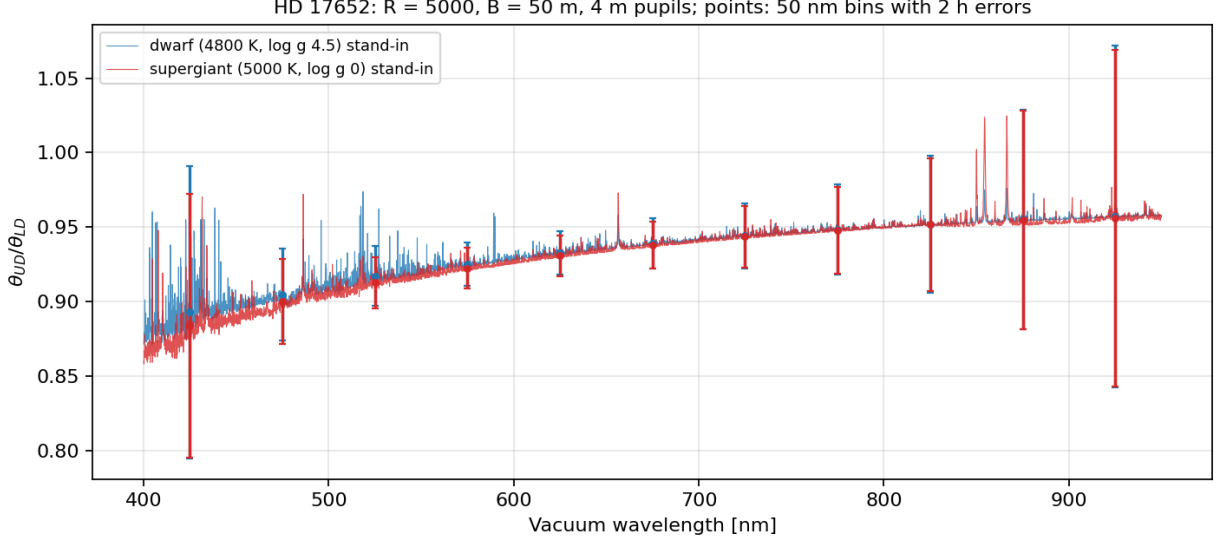


Figure 3: θ_{UD}/θ_{LD} per $R = 5000$ channel for HD 17652 at $B = 50$ m with 4 m pupils, for both stand-in profiles; points are inverse-variance means in 50 nm bins with the 2 h errors of the $R = 5000$ correlator backend. The errors grow to the red because 50 m barely resolves the star there (the optimum baseline is band-dependent); the strong lines (Ca II H and K, the G band, Mg b, H α , the Ca II triplet) stand 2–4% above the continuum.

of a K giant measured rather than assumed, which is the test the authors say G/K atmospheres need.

3 To do

- **ToDo: Request the red-clump models from Hamburg** (P. Hauschildt, E. Baron; Hauschildt is away until 10 October 2026): NewEra HSR-RF at $[M/H] = 0$ for $T_{\text{eff}} = 4700, 4800, 4900$ K $\times \log g = 2.0, 2.5, 3.0$ (nine models), optionally 4800 K / $\log g = 2.5$ at $[M/H] = -0.5$ and $+0.5$. Then: bin at NERSC (`tools/bin_redclump.sh`, both wavelength ranges), re-run `scripts/redclump_ii.py` and `redclump_figures.py`, and replace every stand-in number here. The gravity dependence of $R_{\text{limb}}/R_{\tau=1}$ is the number this note cannot give.
- **ToDo: Match the PIONIER convention.** Establish which radius the Gallenne et al. SATLAS fits report (the SATLAS reference radius vs the visible limb) and evaluate the NewEra giant’s profile with the same definition before quoting a diameter offset.
- **ToDo: Photometry check.** Unanchored, the stand-in models are 0.2 mag brighter than the observed V and H at θ_{LD} (0.34 with the supergiant’s larger drawn disk); the anchoring hides this. See whether the giant model and its metallicity close it, since a 0.2 mag surface-brightness offset at fixed T_{eff} and θ is itself an SBC-relation question.
- **ToDo: H-band detector.** Add an InGaAs-SPAD / SNSPD preset (PDE curve, jitter, dark rate, dead time) so that the H-band case is computed with the same realism as the optical ones.

- **ToDo: Observation realism.** Baseline projection over the 2 h, atmospheric extinction and airmass, and a Monte Carlo of the s fit (`hbtsim.montecarlo`) with the singles-based normalization, to check that the Cramér–Rao bound is attained.
- **ToDo: CTAO case.** Their “666 baselines” with the actual SST layout (`bispectrum.Array`): redundancy, projected baselines over the night, second-lobe coverage across channels.

Code and logs. In the repository `nugent68/binary` (commit `e8a197f` and following):

- `hbtsim/diameter.py`; `hbtsim/single.py` (HD_17652, HD_360)
- `scripts/redclump_ii.py`, `redclump_compare.py`, `redclump_figures.py`
- `output/logs/redclump_ii_dwarf.txt`, `redclump_ii_supergiant.txt`, `redclump_compare.txt`
- `docs/redclump_ii.md` (the same results in Markdown)